American Computer Science League

2023-2024 ● Contest 3: Short Problems Solutions ● Intermediate Division

1. Boolean Algebra

$$\begin{aligned} A\overline{B} + \overline{A\,B} + A &= A\overline{B} + (\overline{A} + \overline{B}) + A \\ &= A\overline{B} + \overline{B} + (\overline{A} + A) \\ &= A\overline{B} + \overline{B} + 1 \\ &= 1 \end{aligned}$$

E. 4

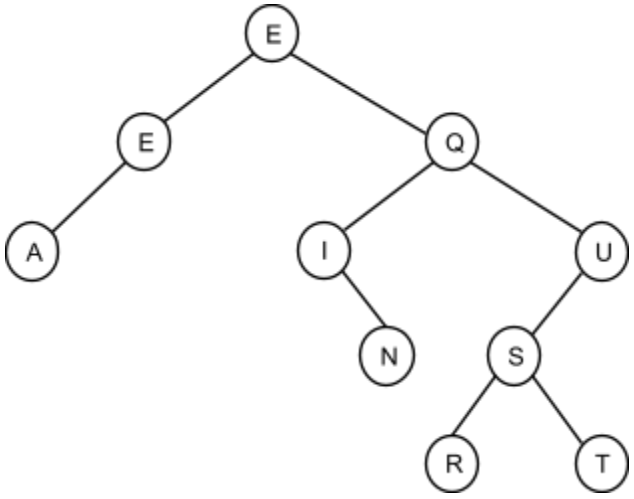
2. Boolean Algebra

$$\begin{aligned} \overline{A\,B} + \overline{A(B + \overline{A})} &= \overline{A} + \overline{B} + \overline{A} + (\overline{B + \overline{A}}) \\ &= \overline{A} + \overline{B} + \overline{A} + \overline{\overline{A}}\,\overline{B} \\ &= \overline{A} + \overline{B} + A\overline{B} \\ &= \overline{A} + \overline{B}\,(1 + A) \\ &= \overline{A} + \overline{B} \end{aligned}$$

A. $\overline{A} + \overline{B}$

3. Data Structures

The binary search tree for EQUESTRIAN is:



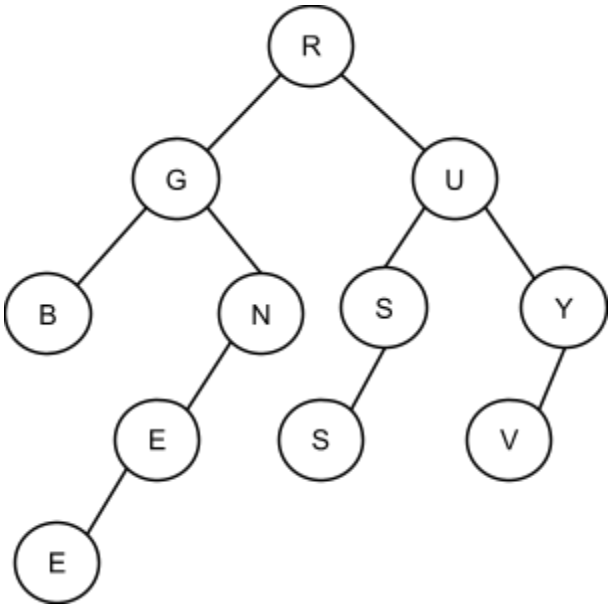
The nodes with only one child are: E, I, U. There are 3.

C. 3

4. Data Structures The queue is constructed using FIFO as follows: F FE FEN EN ENC NC NCI NCIN CIN CING ING NG Items left from the rear to the front of the queue are: G N	D. G N
5. Boolean Algebra $A(\overline{B} + C) + \overline{AB} + C = A\overline{B} + AC + \overline{A} + \overline{B} + C$ $= \overline{B}(A + 1) + C(A + 1) + \overline{A}$ $= \overline{B} + C + \overline{A}$ The only time this expression is FALSE is when all terms are 0, which is (1, 1, 0). So 7 triples make it TRUE.	E. 7
6. Boolean Algebra $\overline{A}B(\overline{C + A}) + A\,B(\overline{\overline{A} + C})$ $= \overline{A}\,B\,\overline{C}\,\overline{A} + A\,B\,A\,\overline{C}$ $= \overline{A}\,B\,\overline{C} + A\,B\,\overline{C}$ $= B\,\overline{C}\,(\overline{A} + A)$ $= B\,\overline{C}$	D. $B\overline{C}$

7. Data Structures

The binary search tree for **RUGBYSEVENS** is:



The nodes at depth 2 from left to right are: B, N, S, Y

B. B, N, S, Y

8. Data Structures

The stack from top to bottom is
A, AR, ARC, AR, ARH, AR, A, AE, AER, AERY, AER, AE.
The next item popped would be E.

C. E

9. FSAs

a. 01111101 satisfies it
b. 011101 satisfies it
c. 0111111 must contain a 0 before the final 1
d. 0111001 satisfies it
e. 0110 must end in 1
f. 011110001 satisfies it

D. 4

10. Regular Expressions

[J-PT-V]*[^c^f-m]*[2468]

a. Jjp2 fails at j in [^c^f-m]
b. KULaazzz24 fails at 4 in [2468]
c. L6 satisfies it
d. TTOOB8 satisfies it
e. dstxyz4 satisfies it
f. 4 satisfies it

C. c, d, e, f